1.2 II order systems

Consider the second order mechanical system represented by spring, mass, and dashpot. We had derived the differential equation governing the dynamics of the system as

$m\ddot{x} + b\dot{x} + kx = F(t)$……………………………(1)

This equation is a second order, ordinary, non-homogeneous, linear differential equation with constant coefficients. Since it is non-homogeneous equation, the solution will have two parts- a homogeneous solution (from the corresponding homogeneous part) and a particular solution (depending upon the nature of the particular type of input F(t)). The homogeneous solution is known as the natural response and the particular solution is known as the forced response.

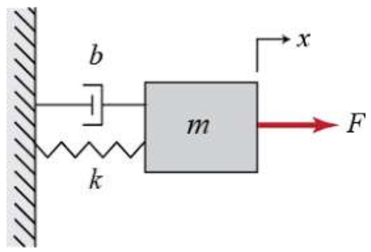

Fig 5. Second order mechanical system

Consider the corresponding homogeneous equation

$m\ddot{x} + b\dot{x} + kx = 0$ …………………………..(2)

Since the equation is linear, any exponential function will be a possible solution. Assuming a solution

$x(t) = Ae^{pt}$ ……………………………………..(3) where A is an arbitrary constant which can be evaluated from the initial conditions and p is a constant but not arbitrary. Substituting (3) in (2), we get

$mp^2Ae^{pt} + bpAe^{pt} + kAe^{pt} = 0$

$Ae^{pt}(mp^2 + bp + k) = 0,$

A or e^{pt} can not be equal to zero as it gives x(t)=0 for all values of time which is a trivial solution. Hence for (3) to be a solution of (2), the condition is

$(mp^2 + bp + k) = 0$ …………………………..(4)

This is known as the characteristic equation, because the solution of it is going to give some particular characteristics of the system (which is p in this case).

$p = \frac{-b \pm \sqrt{b^2 - 4km}}{2m}$ ………………………………….(5)

We have two roots for p. Hence $x_1(t) = A_1e^{p_1t}$ and $x_2(t) = A_2e^{p_2t}$ are possible solutions of (2). Since the equation (2) is of second order, the solution should have two arbitrary constants. Because of the fact that the equation is linear, we can apply the principle of superpostion and any linear combination of the two solutions will be the possible solution.

Hence the homogeneous solution is

$x(t) = A_1e^{p_1t} + A_2e^{p_2t}$ …………………………..(6) if the roots p_1 and p_2 are distinct (real or complex)

$x(t) = (A_1 + A_2t)e^{pt}$ ………………………………(7) if the roots are real and equal. (Refer your S_1 Maths)

Depending on the value of the discriminant $\sqrt{b^2 - 4km}$ of (5), we have three cases

Case(i) $b^2 > 4km$

In this case, the roots are real and distinct. The effect of damping force (b) is more predominant than the spring force (k) and inertia force(m). This system is called an overdamped system.

Case(ii) $b^2 < 4km$

In this case, the roots are complex conjugates. The damping effect is less predominant in this system, which is called an underdamped system.

Case(iii) $b^2 = 4km$

Here, the roots are real and equal. This is called a critically damped system. Of the three cases, this is the only equality expression. We can find out the critical damping coefficient of the system as

$b_c = \sqrt{4km} = 2m\sqrt{\frac{k}{m}} = 2m\omega_n$ ....(8) where $\omega_n = \sqrt{\frac{k}{m}}$. is the undamped natural frequency of the system .

So, for every given system, we can calculate the critical damping coefficient associated with that system because it involves only the system properties m and ω_n (or m and k). But the actual damping of the system can be different.

For example; consider a mass 5 kg which is suspended with a spring of stiffness 1000 N/m. The critical damping coefficient of the system can be calculated as

$b_c = \sqrt{4km} = \sqrt{4\text{x}1000\text{x}5}$=141.4 N/ms^{-1} ($\omega_n = \sqrt{\frac{1000}{5}}$ = 14.14 rad/s, b_c=2x5x14.14=141.4 N/ms^{-1})

Suppose, it is allowed to vibrate in three media viz. air, water, and oil. Then, the actual damping present in the system will be different in these three cases (depending on their viscosity) which may not be equal to the critical damping we have calculated above. The actual damping can be more, equal, or less than the critical damping. Hence, we can always represent the actual damping present in a given system as a factor of the critical damping coefficient of that system (which can be evaluated from the system properties)

$b = \zeta\, b_c$ where ζ(zeta) is known as the damping ratio.

$$\zeta = \frac{actual\ damping\ coefficient\ of\ the\ system}{critical\ damping\ coeffient\ of\ the\ system}$$

Therefore, the actual damping coefficient of the system

$b = \zeta\, b_c = 2m\zeta\omega_n$..................................(9)

If ζ=1, the system is critically damped.

If ζ>1, the system is overdamped.

If ζ<1, the system is underdamped.

Now, (5) can be written as

$$p_{1,2} = \frac{-b}{2m} \pm \frac{\sqrt{b^2-4km}}{2m} = \frac{-b}{2m} \pm \sqrt{\left(\frac{b}{2m}\right)^2 - \left(\frac{k}{m}\right)} = -\zeta\omega_n \pm \sqrt{(\zeta\omega_n)^2 - (\omega_n)^2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1} \quad ...(10)$$

If $\zeta > 1$, the roots p_1 and p_2 are real and distinct. Then the homogenous solution (6) can be written as

$$x(t) = A_1 e^{\left(-\zeta\omega_n + \omega_n\sqrt{\zeta^2-1}\right)t} + A_2 e^{\left(\zeta\omega_n - \omega_n\sqrt{\zeta^2-1}\right)t}$$

$$x(t) = e^{-\zeta\omega_n t}\left(A_1 e^{(\omega_n\sqrt{\zeta^2-1})t} + A_2 e^{(-\omega_n\sqrt{\zeta^2-1})t}\right) \quad ....(11)$$

If $\zeta = 1$, the roots p_1 and p_2 are real and equal ($p_{1,2} = -\omega_n$). Then the homogenous solution **(7)** can be written as

$$x(t) = (A_1 + A_2 t)e^{-\omega_n t} \quad ....(12)$$

If $\zeta < 1$, the roots p_1 and p_2 are complex conjugates.

$p_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2} = -\zeta\omega_n \pm j\omega_d$ where damped frequency $\omega_d = \omega_n\sqrt{1-\zeta^2}$. Then the homogenous solution (6) can be written as

$x(t) = A_1 e^{(-\zeta\omega_n + j\omega_d)t} + A_2 e^{(-\zeta\omega_n - j\omega_d)t}$ $= e^{-\zeta\omega_n t}(A_1 e^{j\omega_d t} + A_2 e^{-j\omega_d t})$............(13a)

$= e^{-\zeta\omega_n t}\{A_1(\cos\omega_d t + j\sin\omega_d t) + A_2(\cos\omega_d t - j\sin\omega_d t)\}$

$= e^{-\zeta\omega_n t}\{(A_1 + A_2)\cos\omega_d t + j(A_1 - A_2)\sin\omega_d t\}$

$= e^{-\zeta\omega_n t}(C_1 \cos\omega_d t + C_2 \sin\omega_d t)$............(13b) C_1 and C_2 are the two new arbitrary constants

$= e^{-\zeta\omega_n t}(X_0 \sin\theta \cos\omega_d t + X_0 \cos\theta \sin\omega_d t)$ where $C_1 = X_0 \sin\theta$ and $C_2 = X_0 \cos\theta$ so that

$X_0 = \sqrt{C_1^2 + C_2^2}$ and $\theta = \tan^{-1}\frac{C_1}{C_2}$

i.e, $x(t) = X_0 e^{-\zeta\omega_n t}\sin(\omega_d t + \theta)$..................(14) where X_0 and θ are the new arbitrary constants.

The response of an underdamped system can be represented by any one of the three equations (13a), (13b), or (14).

Now we have obtained the natural response (homogeneous solution) of a second order system

$x(t) = e^{-\zeta\omega_n t}(A_1 e^{(\omega_n\sqrt{\zeta^2-1})t} + A_2 e^{(-\omega_n\sqrt{\zeta^2-1})t}$ if the system is overdamped ($\zeta > 1$)

$x(t) = (A_1 + A_2 t)e^{-\omega_n t}$ if the system is critically damped ($\zeta = 1$)

$x(t) = X_0 e^{-\zeta\omega_n t}\sin(\omega_d t + \theta)$ if the system is underdamped $\zeta < 1$)

The values of the arbitrary constants have to be determined from the initial conditions.

If we look at the first expression above for the overdamped system, irrespective of the values of the arbitrary constants as $t \to \infty$,

$e^{-\zeta\omega_n t} \to 0$, $e^{(\omega_n\sqrt{\zeta^2-1})t} \to \infty$, $e^{(-\omega_n\sqrt{\zeta^2-1})t} \to 0$

hence x(t) $\to 0$

In second expression of critical damped system as $t \to \infty$,

$(A_1 + A_2 t) \to \infty$, $e^{-\omega_n t} \to 0$, hence x(t) $\to 0$

In both the above cases, the system will not oscillate. Once the vibration is initiated by giving some initial conditions, the system comes to equilibrium without oscillation. The time taken by the system to come to equilibrium is smaller for critically damped system than an overdamped.

For the under damped system, the response is a sine curve with amplitude $X_0 e^{-\zeta\omega_n t}$ and circular frequency ω_d. The period of oscillation $T = \frac{2\pi}{\omega_d}$. The amplitude of oscillation $X_0 e^{-\zeta\omega_n t}$ decays exponentially to zero as $t \to \infty$.

Only an underdamped sytem will vibrate/oscillate when it is disturbed from equilibrium.

The homogeneous solutions for the three different cases are plotted in figure 6. The arbitrary constants are assumed as A_1=1, A_2=0, X_0=1, and $\theta=90^0$ and ω_n=5

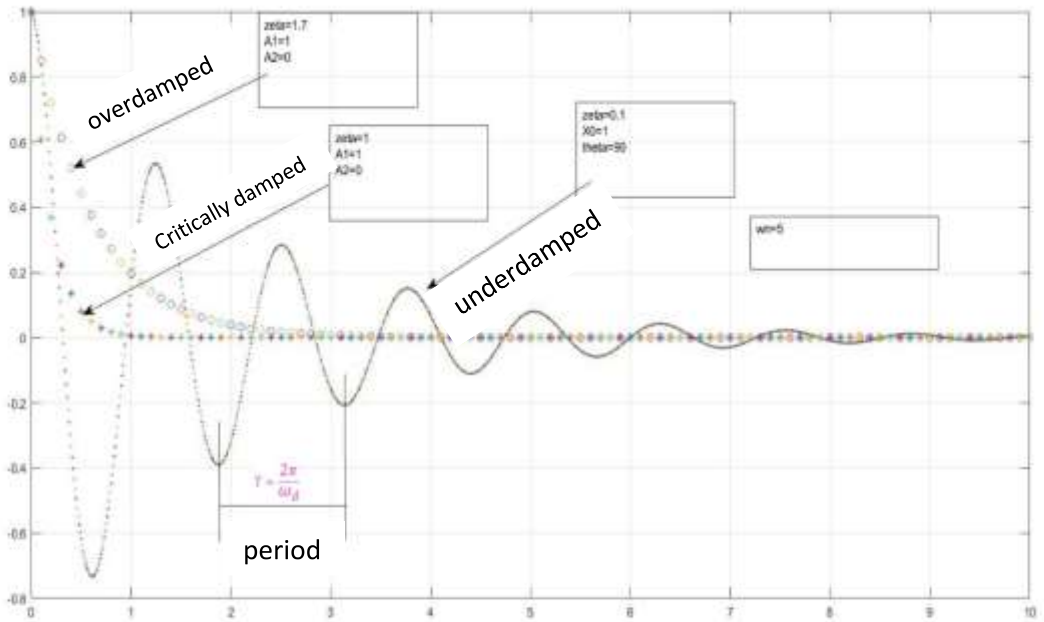

Fig. 6 Natural responses of a second order mechanical system for different values of

$$\zeta = 0.1, 1, 1.7$$

We can see that irrespctive of the values of damping factor, the natural response vanishes after some time. Hence this is known as transient response.

The particular solution is obtained by considering a particular input F(t). The response that we get is known as the forced response which will exist as long as F(t) exists. We will consider the forced responses for step and impulse inputs later. The complete solution (total response) will be the sum of natural response(homogeneous solution) and forced response (particular solution)

#Exercise 5

Find the values of the arbitrary constants in the expressions for the homogeneous responses of overdamped ($\zeta = 1.7$), critically damped ($\zeta = 1$), and underdamped ($\zeta = 0.7$) with undamped natural frequency ($\boldsymbol{\omega_n = 5\ rad/s}$) for the following set of initial conditions

(i) $x(0) = 1,\ \dot{x}(0) = 0$ (ii) $x(0) = 0,\ \dot{x}(0) = 1$ (iii) $x(0) = 1,\ \dot{x}(0) = 1$

Plot the responses using Matlab/Scilab/Excel

(Hint: For substituting the velocity initial condition, you need to differentiate the expression for x(t) with respect to time.)

Let us again consider the roots of the characteristic equation given in (10)

$$p_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$

Let us vary the value of ζ keeping ω_n as constant (=5 rad/s)

ζ	$-\zeta\omega_n$	$\omega_n\sqrt{\zeta^2 - 1}$	$p_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$	$p_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$
0	0	j5.00	+j5.0	-j5.0
0.2	-1.0	j4.89	-1.0+j4.89	-1.0-j4.89
0.4	-2.0	j4.58	-2.0+j4.58	-2.0-j4.58
0.6	-3.0	j4.00	-3.0+j4.00	-3.0-j4.00
0.8	-4.0	j3.00	-4.0+j3.00	-4.0-j3.00
1.0	-5.0	0	-5.0	-5.0
1.2	-6.0	3.31	-2.69	-9.31
1.4	-7.0	4.89	-2.11	-11.89
1.6	-8.0	6.25	-1.75	-14.25

Look at the values of p_1 and p_2 in the above table. It gives the values of the roots of the characteristic equation when ζ is varied from 0. The values of the roots are plotted in complex plane in figure 7. The horizontal axis is the real axis and vertical one is the imaginary axis. When $\zeta = 0$ (undamped), the roots are $\pm j5$ which are on the imaginary axis. As we increase the value of ζ to 0.2 (underdamped), the roots are complex conjugates with negative real part and shift to the left side of the imaginary axis. The values of root p_1 are marked in red dots and p_2 are shown in blue dots. When ζ =1 (critically damped), both the roots are equal at -5. When ζ increases above 1(overdamped), both the roots are negative real numbers. One moves to a large negative value and the other approaching to zero. Please note that the value of p_1 will never be equal to zero however high the value of ζ may be, because the negative part $-\zeta\omega_n$ will be always greater than the positive part $\omega_n\sqrt{\zeta^2 - 1}$. **If we continuously change the parameter ζ and plot the roots of the charcteristic equation, we will get the locus of the roots of the characteristic equation which is known as the ROOT LOCUS.** We will discuss the root locus technique in detail later.

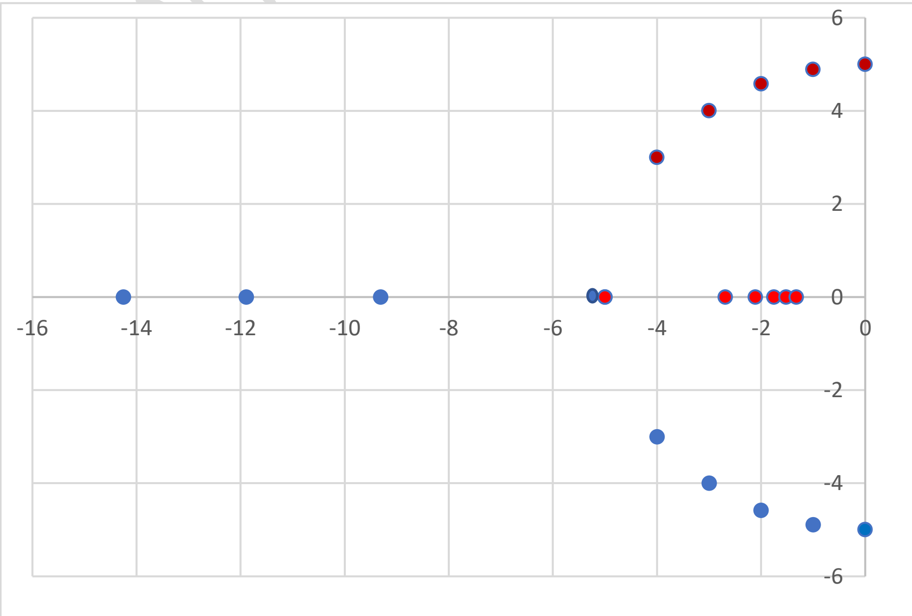

Fig. 7 Roots of the characteristic equation as ζ increases from 0.